Inference with weakly dependent data I

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Plan for this slide deck

1. Standard errors for general time series regression
2. CLT for weakly dependent data
3. Heteroskedasticity and autocorrelation consistent (HAC) estimator, Newey-West SE
4. Application: testing forecast rationality

Time series regression

- Given jointly covariance stationary time series $\{y_t, x_t\}$, consider an OLS regression of y_t on x_t:

$$y_t = \hat{\beta}_0 + \hat{\beta}_1 x_t + \hat{v}_t.$$

- How should we compute standard errors for the coefficients?
- In our discussion of AR models, we have seen that if

$$y_t = \beta_0 + \beta_1 x_t + v_t,$$

  and $\{v_t\}$ is a martingale difference sequence, then we can use the usual SE formula for i.i.d. regressions.
- But in many cases, we may expect v_t to be serially correlated (and therefore certainly not a MDS).

The Phillips curve: Corr(inflation, unemployment) < 0

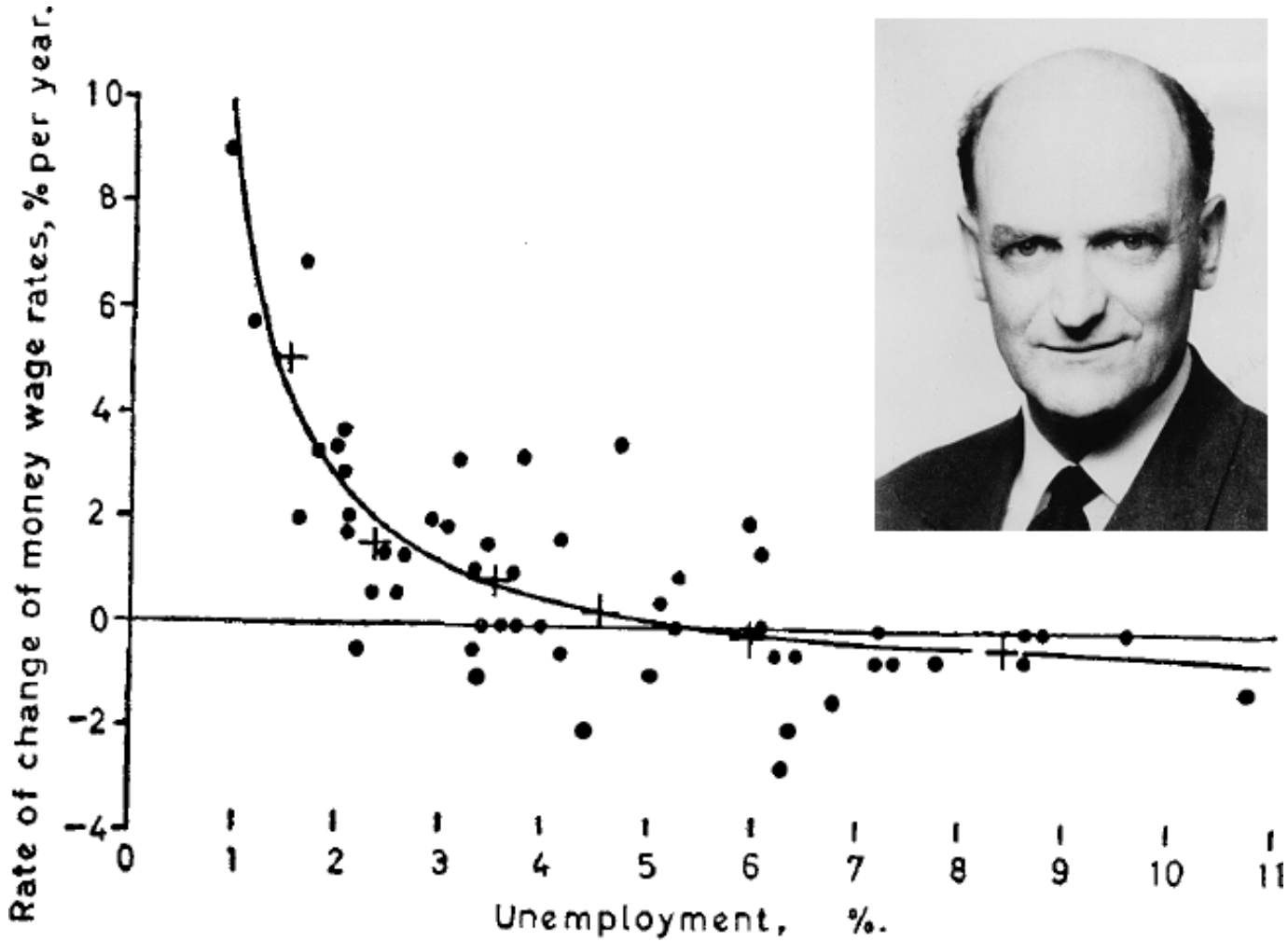
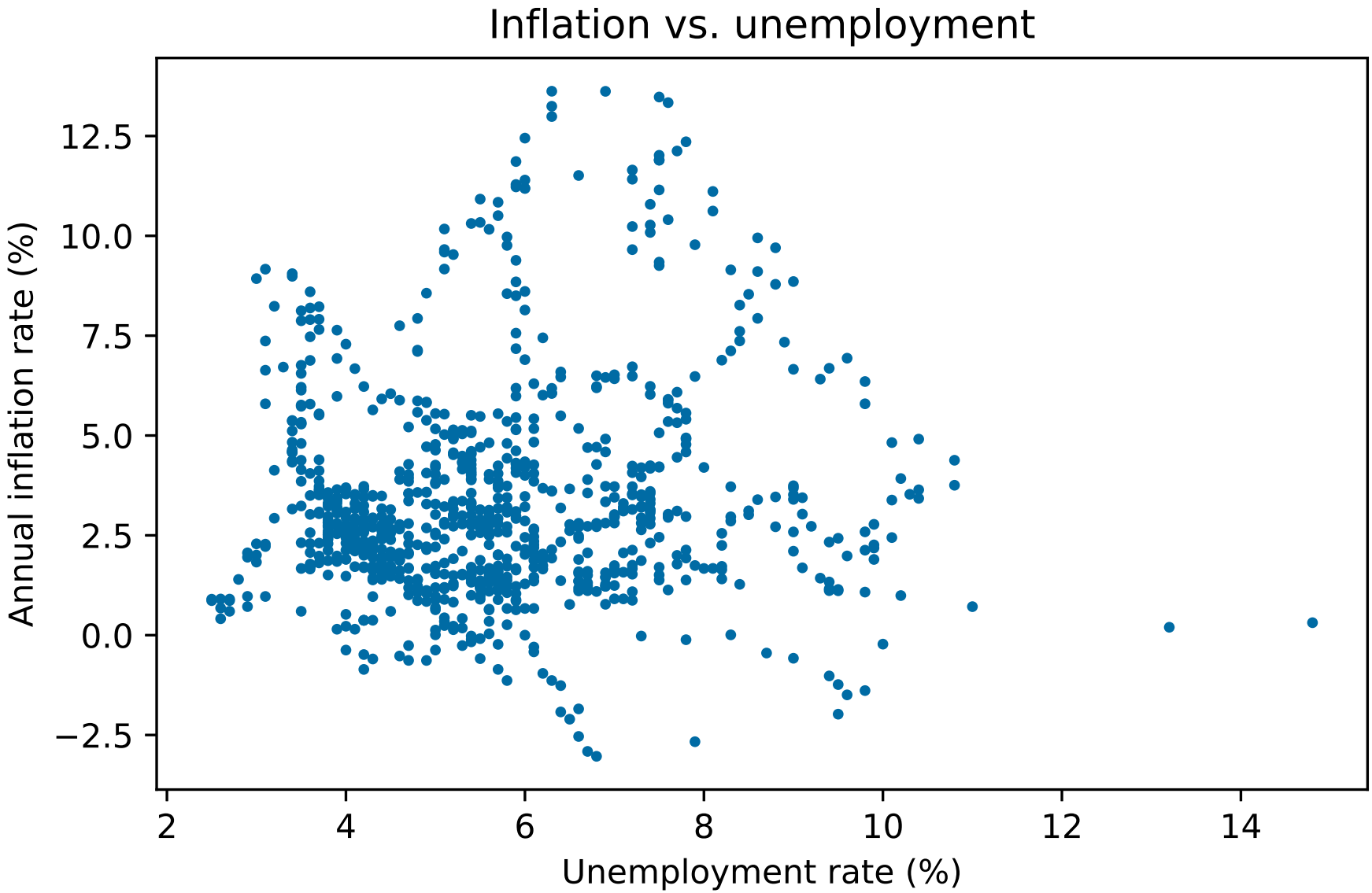


Fig.1. 1861 – 1913

Note: UK data. Source: Phillips (1958)



U.S. data 1948–2026

Inflation: log change in CPI relative to 12 months ago

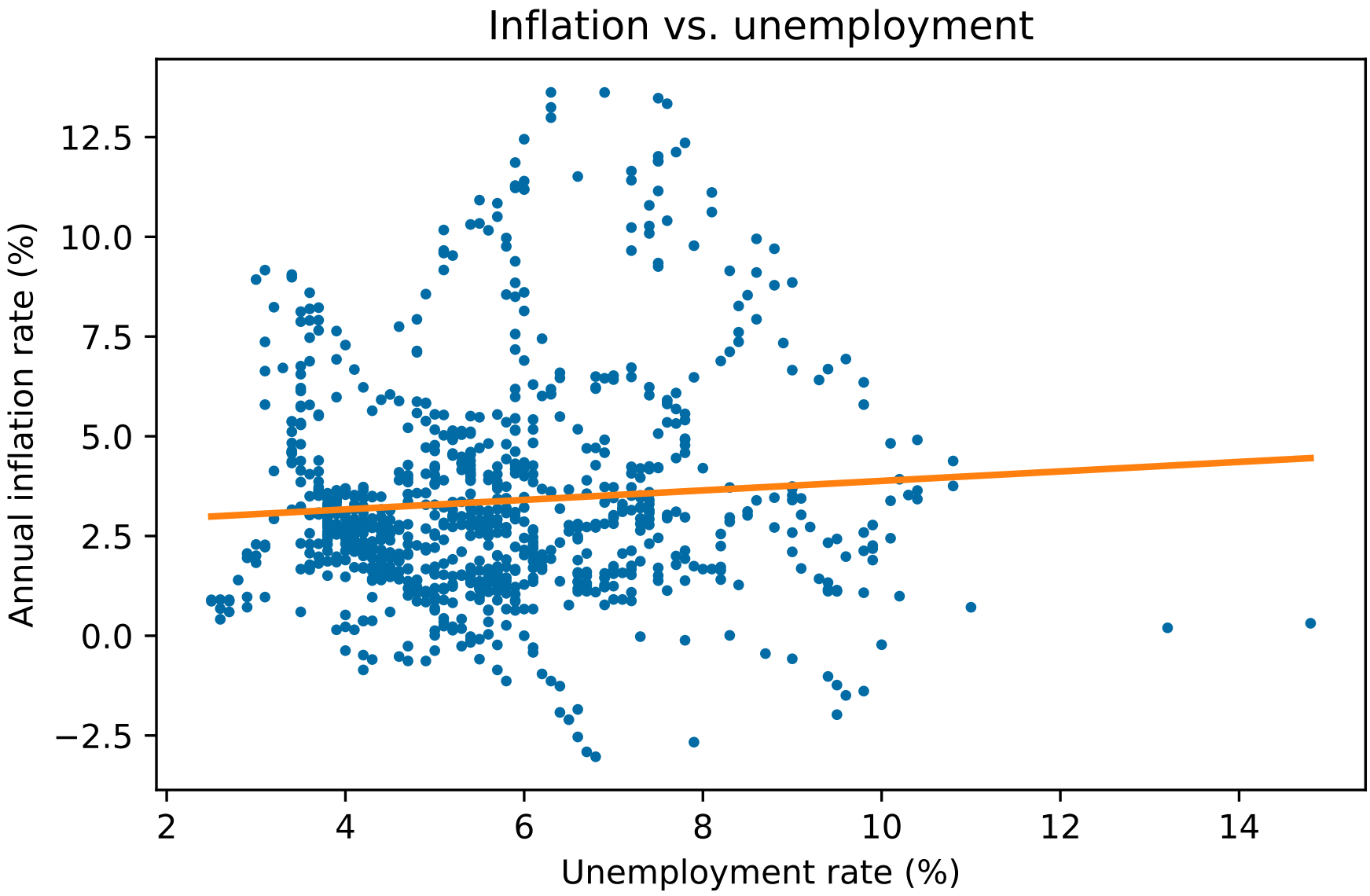
OLS Regression Results

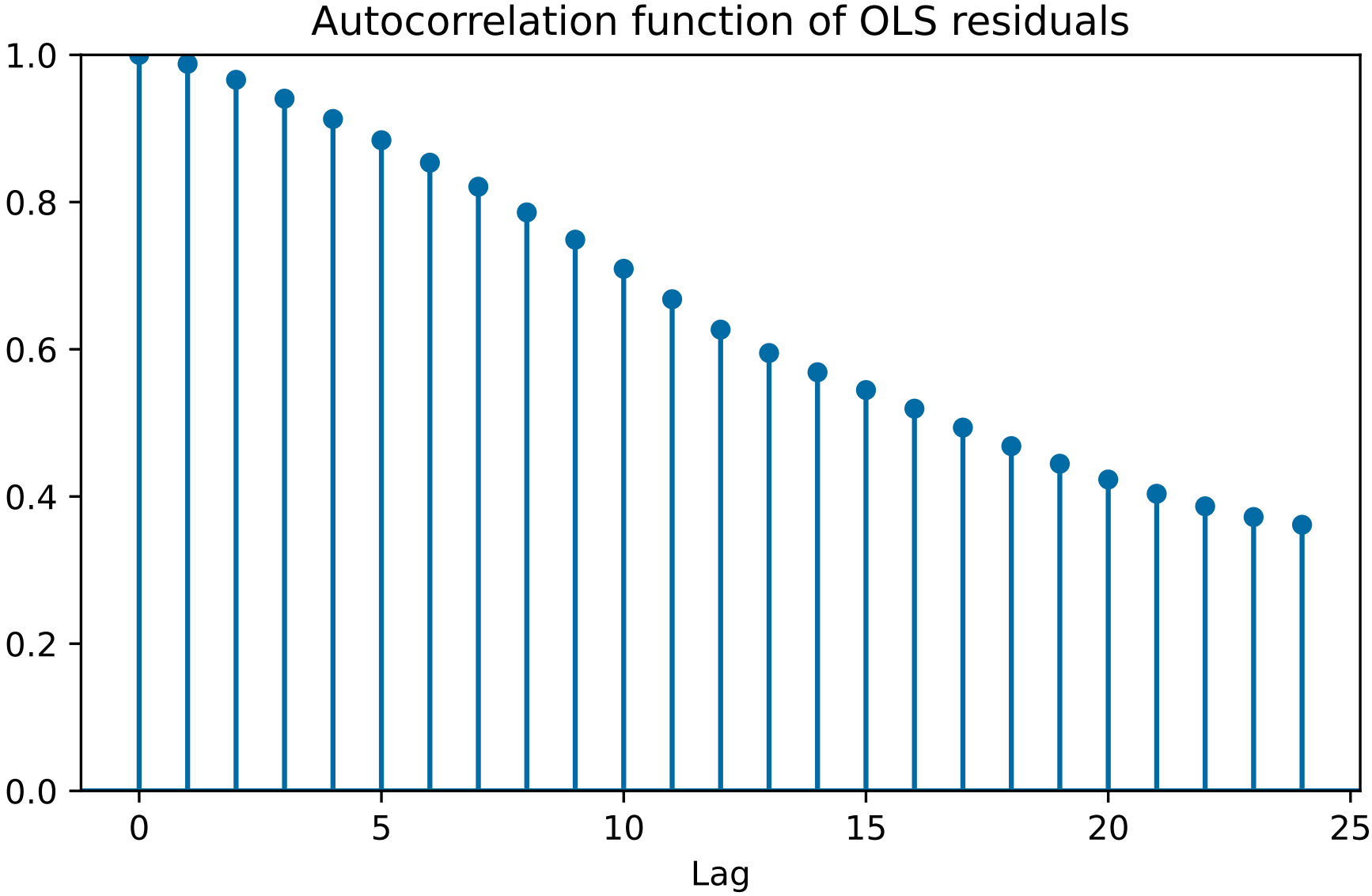
Dep. Variable:	inflation	R-squared:	0.006
Model:	OLS	Adj. R-squared:	0.005
Method:	Least Squares	F-statistic:	4.746
Date:	Mon, 13 Apr 2026	Prob (F-statistic):	0.0296
Time:	20:22:43	Log-Likelihood:	-2227.8
No. Observations:	926	AIC:	4460.
Df Residuals:	924	BIC:	4469.
Df Model:	1		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	2.6917	0.295	9.119	0.000	2.113	3.270
UNRATE	0.1190	0.055	2.178	0.029	0.012	0.226

Omnibus:	195.796	Durbin-Watson:	0.024
Prob(Omnibus):	0.000	Jarque-Bera (JB):	379.007
Skew:	1.228	Prob(JB):	5.01e-83
Kurtosis:	4.947	Cond. No.	21.3

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)


Inflation vs. unemployment
Annual inflation rate (%)
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
2
4
6
8
10
12
14
Unemployment rate (%)



$\{\hat{v}_t\}$ definitely doesn't look like white noise!

Distribution of OLS estimator

- Drop the intercept for now and assume the linear model

$$y_t = \beta x_t + v_t, \quad E[x_t v_t] = 0.$$

  - Not really an assumption. Can always define the **best linear prediction** coefficient

$$\beta \equiv \underset{b}{\operatorname{argmin}} E\left[(y_t - x_t b)^2\right], \quad \text{and} \quad v_t \equiv y_t - \beta x_t.$$

  Then the FOC implies $E[x_t v_t] = 0$.

- OLS estimator of β:

$$\hat{\beta} \equiv \frac{\sum_{t=1}^T y_t x_t}{\sum_{t=1}^T x_t^2} = \beta + \frac{\sum_{t=1}^T v_t x_t}{\sum_{t=1}^T x_t^2}$$

$$\Longrightarrow \sqrt{T}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{T}} \sum_{t=1}^T v_t x_t}{\frac{1}{T} \sum_{t=1}^T x_t^2}.$$

Distribution of OLS estimator (cont.)

$$\sqrt{T}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} v_t x_t}{\frac{1}{T} \sum_{t=1}^{T} x_t^2} \qquad (\dagger)$$

- It turns out that the denominator in $(\dagger)$ satisfies the law of large numbers (LLN) provided that $\{x_t\}$ is stationary and $\lim_{\ell \to \infty} \mathrm{Cov}(x_t^2, x_{t-\ell}^2) = 0$:

$$\frac{1}{T} \sum_{t=1}^{T} x_t^2 \xrightarrow{p} E[x_t^2] \quad \text{as} \quad T \to \infty.$$

- The numerator in $(\dagger)$ is a scaled sample average of terms with mean zero (by assumption). Does it satisfy a central limit theorem (CLT)?

CLT with dependent data

- A simple example shows that the CLT can indeed hold (in modified form) for dependent data.
- Consider the MA(1) process

$$y_t = u_t + \theta u_{t-1}, \quad u_t \overset{iid}{\sim} (0, \sigma^2).$$

- Then

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} y_t = \frac{1}{\sqrt{T}} \Big(u_1 + \theta u_0 + u_2 + \theta u_1 + u_3 + \theta u_2 + \cdots + u_T + \theta u_{T-1} \Big)$$

$$= (1+\theta) \underbrace{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} u_t}_{\overset{d}{\to} N(0,\sigma^2)} + \underbrace{\frac{1}{\sqrt{T}} \theta (u_0 - u_T)}_{\overset{p}{\to} 0} \overset{d}{\to} \boxed{N\Big(0, (1+\theta)^2 \sigma^2\Big)},$$

using the usual CLT for i.i.d. variables (the u_t's) and Slutsky's thm.

CLT with dependent data (cont.)

$$y_t = u_t + \theta u_{t-1}, \quad u_t \overset{iid}{\sim} (0, \sigma^2)$$

- In this example, the asymptotic variance of the scaled sample average is

  $$(1+\theta)^2\sigma^2,$$

  even though

  $$\text{Var}(y_t) = (1+\theta^2)\sigma^2.$$

  (Recall the formula for the variance of an MA process.)

- So something is different when applying the CLT to dependent processes, relative to the i.i.d. case: we still get asymptotic normality, but the variance is different.

Variance of sample mean

- Let's derive the variance of the scaled sample mean of a general covariance stationary process $\{y_t\}$ with autocovariances $\gamma(\ell)$.
- Then

$$\text{Var}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} y_t\right) = \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}\text{Cov}(y_t, y_s)$$
$$= \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}\gamma(t-s) = \gamma(0) + 2\sum_{\ell=1}^{T-1}\frac{T-\ell}{T}\gamma(\ell).$$

  - Exercise: Make sure you understand each step.

Variance of sample mean (cont.)

$$\text{Var}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} y_t\right) = \gamma(0) + 2\sum_{\ell=1}^{T-1}\frac{T-\ell}{T}\gamma(\ell)$$

- Assume that the process $\{y_t\}$ is **weakly dependent** in the sense that

$$\sum_{\ell=1}^{\infty}|\gamma(\ell)| < \infty.$$

  - Requires $\gamma(\ell) \to 0$ sufficiently fast as $\ell \to \infty$.

- Then the "dominated convergence theorem" implies

$$\boxed{\lim_{T\to\infty}\text{Var}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} y_t\right) = \gamma(0) + 2\sum_{\ell=1}^{\infty}\gamma(\ell)}.$$

Long-run variance

$$\lim_{T\to\infty} \text{Var}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} y_t\right) = \gamma(0) + 2\sum_{\ell=1}^{\infty}\gamma(\ell) \equiv \text{LRV}(y_t)$$

- LRV: **long-run variance**.
- If the process is white noise, then $\text{LRV}(y_t) = \gamma(0) = \text{Var}(y_t)$. But otherwise, we generally have $\text{LRV}(y_t) \neq \text{Var}(y_t)$.
- Exercise: show that for a cov. stationary AR(1) process,

$$\text{LRV}(y_t) = \frac{\sigma^2}{(1-\rho)^2} \neq \frac{\sigma^2}{1-\rho^2} = \text{Var}(y_t).$$

CLT with dependent data: result

- Result: if the process $\{y_t\}$ is stationary, has mean zero, is weakly dependent, and satisfies some other regularity conditions, then

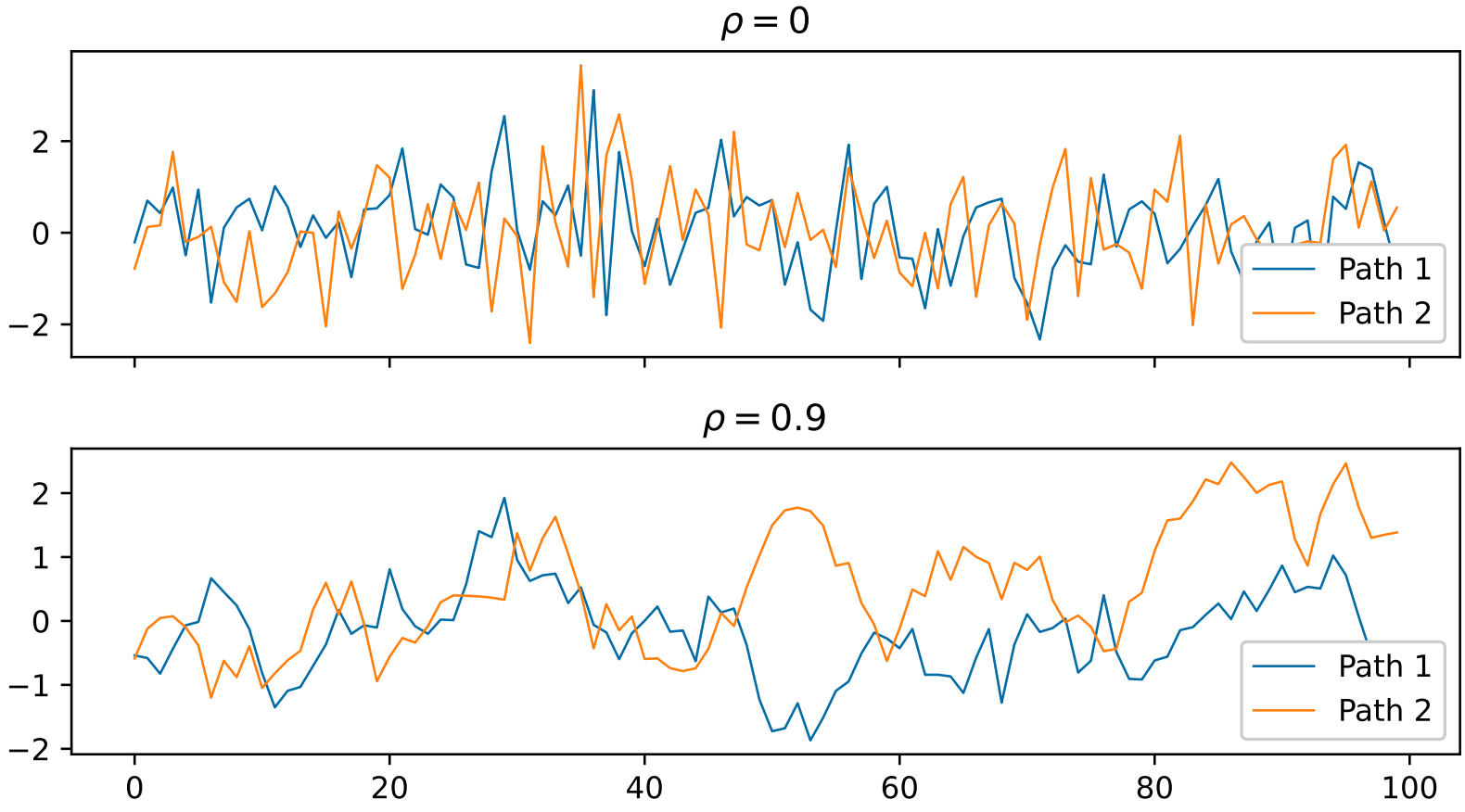
$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} y_t \xrightarrow{d} N\Big(0, \text{LRV}(y_t)\Big).$$

  - Formal results: Hayashi (2000) textbook, chapter 6.5.

- If the process is positively serially correlated, so that $\gamma(\ell) \geq 0$ for $\ell \geq 1$, then $\text{LRV}(y_t) \geq \text{Var}(y_t)$.

- So compared with the case of i.i.d. data, the sample average of a positively serially correlated time series has higher variance.

  - It's effectively as if the sample size is smaller: we observe fewer (nearly) independent observations.



- The plot shows two AR(1) processes with different persistence ρ, each normalized to have variance $= 1$.
- The white noise process (top) averages out the noise/shocks much faster than the persistent process (bottom).

Long-run variance: linear filter

- Result: suppose $y_t = b(L)x_t$ for some (absolutely summable) linear filter $b(L) \equiv \sum_{\ell=-\infty}^{\infty} b_\ell L^\ell$. Then

$$\text{LRV}(y_t) = [b(1)]^2 \times \text{LRV}(x_t).$$

  - Recall: $b(1) \equiv \sum_{\ell=-\infty}^{\infty} b_\ell$.

- Exercise: show that the MA(1) process $y_t = u_t + \theta u_{t-1}$, with $u_t \sim WN(\sigma^2)$, has $\text{LRV}(y_t) = (1+\theta)^2\sigma^2$.

  - Consistent with our earlier expression for the asymptotic variance of the sample average of this process.

Distribution of OLS estimator: conclusion

- Back to the OLS estimator:

$$\sqrt{T}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} v_t x_t}{\frac{1}{T} \sum_{t=1}^{T} x_t^2}.$$

- We have seen that, under some weak conditions,

$$\frac{1}{T} \sum_{t=1}^{T} x_t^2 \xrightarrow{p} E[x_t^2], \quad \text{and} \quad \frac{1}{\sqrt{T}} \sum_{t=1}^{T} v_t x_t \xrightarrow{d} N\Big(0, \mathrm{LRV}(v_t x_t)\Big).$$

- Hence, by Slutsky's theorem,

$$\boxed{\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} N\left(0, \frac{\mathrm{LRV}(v_t x_t)}{\left(E[x_t^2]\right)^2}\right).}$$

Standard errors for time series regression

- Written a bit more loosely, when T is large,

$$\hat{\beta} \overset{approx}{\sim} N\left(\beta, \frac{1}{T} \times \frac{\text{LRV}(v_t x_t)}{\left(E[x_t^2]\right)^2}\right).$$

- For this to be useful for inference, we need to estimate the variance.
- Suppose we had a consistent estimator $\widehat{\text{LRV}}$ of $\text{LRV}(v_t x_t)$. Then we could use the standard errors

$$SE(\hat{\beta}) \equiv \sqrt{\frac{1}{T} \times \frac{\widehat{\text{LRV}}}{\left(\frac{1}{T}\sum_{t=1}^{T} x_t^2\right)^2}}.$$

- How to consistently estimate the LRV?

Long-run variance: estimation

$$\text{LRV}(y_t) \equiv \sum_{\ell=-\infty}^{\infty} \gamma(\ell) = \gamma(0) + 2\sum_{\ell=1}^{\infty} \gamma(\ell)$$

- Most natural estimator:

$$\widetilde{LRV} \equiv \hat{\gamma}(0) + 2\sum_{\ell=1}^{T-1} \hat{\gamma}(\ell),$$

  using the sample autocovariances

$$\hat{\gamma}(\ell) \equiv \frac{1}{T} \sum_{t=\ell+1}^{T} (y_t - \bar{y})(y_{t-\ell} - \bar{y}), \quad \ell = 0, 1, \ldots, T-1.$$

- But this estimator turns out to be *inconsistent*! (In fact, it equals 0.)
- Intuitively, $\hat{\gamma}(\ell)$ is a poor estimator of $\gamma(\ell)$ for $\ell \approx T$.

Long-run variance: estimation (cont.)

- Consider instead the weighted estimator

$$\widehat{LRV} \equiv \hat{\gamma}(0) + 2\sum_{\ell=1}^{B} K\left(\frac{\ell}{B}\right)\hat{\gamma}(\ell),$$

  where $K : [0,1] \to [-1,1]$ is a function called the **kernel**, and B is an integer called the **bandwidth**. We require $K(0) = 1$.

  - $\widehat{LRV}$ sums over a smaller number B of autocovariances. Can use the kernel to down-weight the noisy estimates at longer lags $\ell \approx B$.

- Result: if $B = B(T)$ is a function of T such that $B \to \infty$ and $B/T \to 0$ (and some other regularity conditions hold), then $\widehat{LRV}$ is consistent.

- We call $\widehat{LRV}$ a **heteroskedasticity and autocorrelation consistent** (HAC) estimator.

Long-run variance: Newey-West estimator

- By far the most popular HAC estimator is the **Newey-West** estimator. It sets $K(x) = 1 - x$ (the **Bartlett kernel**):

$$\widehat{LRV}_{\text{NW}} \equiv \hat{\gamma}(0) + 2 \sum_{\ell=1}^{B-1} \left(1 - \frac{\ell}{B}\right) \hat{\gamma}(\ell),$$

- Rule of thumb bandwidth from Stock & Watson textbook: $B = 0.75 \times T^{1/3}$ (rounded up).

- To compute OLS standard errors, we need LRV$(x_t v_t)$. Estimate this by applying the Newey-West formula to the sample autocovariances of the series $\{x_t \hat{v}_t\}$, where $\hat{v}_t$ are the OLS residuals.

```
# OLS regression with Newey-West SE
T_nw = model.nobs
nw_bw = int(np.ceil(0.75 * T_nw**(1/3)))
model_nw = sm.OLS(df['inflation'], X).fit(cov_type='HAC',
    cov_kwds={'maxlags': nw_bw-1})
print(model_nw.summary())
```

Note: `maxlags` $= B - 1$

OLS Regression Results

Dep. Variable:	inflation	R-squared:	0.006
Model:	OLS	Adj. R-squared:	0.005
Method:	Least Squares	F-statistic:	0.7345
Date:	Mon, 13 Apr 2026	Prob (F-statistic):	0.392
Time:	20:22:43	Log-Likelihood:	-2227.8
No. Observations:	926	AIC:	4460.
Df Residuals:	924	BIC:	4469.
Df Model:	1		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	2.6917	0.743	3.622	0.000	1.235	4.148
UNRATE	0.1190	0.139	0.857	0.391	-0.153	0.391

Omnibus:	195.796	Durbin-Watson:	0.024
Prob(Omnibus):	0.000	Jarque-Bera (JB):	379.007
Skew:	1.228	Prob(JB):	5.01e-83
Kurtosis:	4.947	Cond. No.	21.3

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 7 lags and without small sample correction

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2.5 × larger than non-HAC SE!

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 7 lags and without small sample correction

Extensions

- Straight-forward generalization to multivariate regression:

$$y_t = \hat{\beta}_0 + \hat{\beta}_1 x_{1t} + \cdots + \hat{\beta}_k x_{kt} + \hat{v}_t.$$

  - CLT and HAC estimator can be extended to vector processes.
  - F-test for time series data.
  - Most statistics software can do all this for you.

- HAC estimation typically does not work well in small samples when the data is highly persistent. Can ameliorate this by using a larger bandwidth B and adjusting the critical value of our tests/CIs.
  - Review of state of the art: Lazarus, Lewis, Stock & Watson (2018).

OLS Regression Results

Dep. Variable:	inflation	R-squared:	0.235
Model:	OLS	Adj. R-squared:	0.233
Method:	Least Squares	F-statistic:	17.19
Date:	Mon, 13 Apr 2026	Prob (F-statistic):	4.70e-08
Time:	20:22:43	Log-Likelihood:	-2106.7
No. Observations:	926	AIC:	4219.
Df Residuals:	923	BIC:	4234.
Df Model:	2		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.9934	0.737	2.705	0.007	0.549	3.438
UNRATE	0.2086	0.136	1.536	0.124	-0.057	0.475
oil_change	0.0454	0.008	5.822	0.000	0.030	0.061

Omnibus:	104.306	Durbin-Watson:	0.047
Prob(Omnibus):	0.000	Jarque-Bera (JB):	146.152
Skew:	0.837	Prob(JB):	1.83e-32
Kurtosis:	3.992	Cond. No.	103.

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 7 lags and without small sample correction

Another application: testing forecast rationality

- Mankiw, Reis & Wolfers (2003) test (among other things) whether forecasts of inflation π_t are rational.
- Forecast sources:
  - Michigan survey of households
  - Livingston and SPF surveys of professional forecasters
- Regress 12-month-ahead forecast errors $\pi_t - \pi_{t|t-12}$ on various time series to test if the errors can be systematically predicted.
- The multi-step forecast errors must be serially correlated (even if forecasts were rational), since they depend on all shocks that hit the economy between time $t-12$ and t.

  $\Longrightarrow$ Need Newey-West SE.

Table 4 TESTS OF FORECAST RATIONALITY: MEDIAN INFLATION EXPECTATIONS[1]

	Michigan	*Michigan-experimental*	*Livingston*	*SPF (GDP deflator)*
Panel A: testing for bias: $\pi_t - E_{t-12}\,\pi_t = \alpha$				
α: mean error	0.42%	–0.09%	0.63%**	–0.02%
(Constant only)	(0.29)	(0.34)	(0.30)	(0.29)
Panel B: Is information in the forecast fully exploited? $\pi_t - E_{t-12}\,\pi_t = \alpha + \beta\, E_{t-12}\pi_t$				
β: $E_{t-12}\,[\pi_t]$	0.349**	–0.060	0.011	0.026
	(.161)	(.207)	(.142)	(.128)
α: constant	–1.016%*	–0.182%	0.595%	–0.132%
	(.534)	(.721)	(.371)	(.530)
Adj. R^2	0.197	–0.003	–0.011	–0.007
Reject eff.? $\alpha = \beta = 0$	Yes	No	Yes	No
(p-value)	(p = 0.088)	(p = 0.956)	(p = 0.028)	(p = 0.969)
Panel C: Are forecasting errors persistent? $\pi_t - E_{t-12}\,\pi_t = \alpha + \beta\,(\pi_{t-12} - E_{t-24}\pi_{t-12})$				
β: $\pi_{t-12} - E_{t-24}\,[\pi_{t-12}]$	0.371**	.580***	0.490***	0.640***
	(0.158)	(0.115)	(0.132)	(0.224)
α: constant	0.096%	0.005%	0.302%	–0.032%
	(0.183)	(0.239)	(0.210)	(0.223)
Adj. R^2	0.164	0.334	0.231	0.375

Source: Mankiw, Reis & Wolfers (2003)

Panel D: Are macroeconomic data fully exploited? $\pi_t - E_{t-12}\pi_t = \alpha + \beta\ E_{t-12}\ [\pi_t] + \gamma\ \pi_{t-13} + \kappa\ i_{t-13} + \delta\ U_{t-13}$

α: constant	–0.816%	0.242%	4.424%***	3.566%***
	(0.975)	(1.143)	(0.985)	(0.970)
β: $E_{t-12}\ [\pi_t]$	0.801***	–0.554***	0.295	0.287
	(0.257)	(0.165)	(0.283)	(0.308)
γ: inflation$_{t-13}$	–0.218*	0.610***	0.205	0.200
	(0.121)	(0.106)	(0.145)	(0.190)
κ: Treasury bill$_{t-13}$	–0.165**	–0.024	–0.319***	–0.321***
	(0.085)	(0.102)	(0.106)	(0.079)
δ: unemployment$_{t-13}$	0.017	–0.063	–0.675***	–0.593***
	(0.126)	(0.156)	(0.175)	(0.150)
Reject eff.? $\gamma = \kappa = \delta = 0$	Yes	Yes	Yes	Yes
(p-value)	(p = 0.049)	(p = 0.000)	(p = 0.000)	(p = 0.000)
Adjusted R^2	0.293	0.382	0.306	0.407
Sample	Nov. 1974– May 2002	1954, Q4– 2002, Q1	1954, H1– 2001, H2	1969, Q4– 2002, Q1
Periodicity	Monthly	Quarterly	Semiannual	Quarterly
N	290	169	96	125

1. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels, respectively (Newey-West standard errors in parentheses; correcting for autocorrelation up to one year).

Source: Mankiw, Reis & Wolfers (2003)

Data frequency and efficiency

- Suppose we've run a regression on quarterly data. Then we realize that the series are also available at monthly frequency over the same time span.
- Using the monthly data, our sample size T triples. Naively, we might expect our OLS SE to decrease by a factor of $\sqrt{3} \approx 1.73$.
- But in most cases in practice, the monthly data will be much more *persistent* than the quarterly data: the relevant LRV is large.
- Hence, our OLS SE usually do not decrease that much when switching from quarterly to monthly, despite the larger T.
- Of course, monthly data is still useful for analyzing high-frequency dynamics.

What did we learn?

- LLN + CLT for dependent data
- Asymptotic distribution of OLS estimator
- Long-run variance
  - $>$ Var for positively serially correlated series
  - Formula for linear filters
  - HAC estimation, Newey-West
- SE for time series regression

Looking ahead

- Instrumental variables regression with time series data
- How can we identify whether economic agents are forward-looking or backward-looking?
  - New Keynesian Phillips curve
  - Consumption Euler equation